

Wearable Technologies for Detecting Heart Failure or Infection 30 Days Post- Aortic Valve Replacement

STEM Buddies Canada

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Abstract:

Aortic valve replacement (AVR) is a common procedure in older adults, but complications like infection or heart failure remain a risk. Current follow-up care depends on patient-reported symptoms and scheduled visits, which may delay early detection. This study explores whether wearable technology, specifically the Fitbit Sense, can identify early signs of complications within 30 days after surgeryⁱ. A total of 384 AVR patients will be randomly assigned to either standard care or continuous monitoring with Fitbit Sense, which tracks heart rate, heart rate variability, rhythm, and skin temperatureⁱⁱ. Baseline vitals will be recorded pre-surgery, and an algorithm will monitor for deviations to alert clinicians. All participants will have weekly check-ins and a final evaluation on day 30, with complications confirmed through clinical tests. Early research suggests physiological changes may occur before symptoms arise. If successful, this study could support wearables as a low-cost, non-invasive tool to improve recovery, reduce readmissions, and enhance patient safety in AVR careⁱⁱⁱ.

Keywords:

Aortic valve replacement (AVR), aortic valve, heart rate variability, photoplethysmography, night signal, inclusion/exclusion, criteria, secondary outcomes

Introduction:

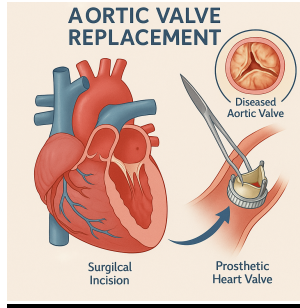


FIGURE 5

Artificial aortic valves are used to replace damaged ones during AVR surgery, most often in older adults. While the surgery can be successful, risks like infection or heart failure remain high, and traditional follow-ups based on symptoms and scheduled visits may miss early warning signs. Wearable devices like the Fitbit Sense use sensors to track heart rate, variability, and skin temperature, potentially detecting subtle physiological changes before symptoms appear^{iv}. This study explores whether continuous Fitbit monitoring during the first 30 days after AVR can identify early signs of complications^v. We compare outcomes between patients receiving standard care and those using wearable monitoring to assess whether this technology can improve recovery and reduce hospital readmissions^{vi}.

Hypothesis:

If patients recovering from aortic valve replacement surgery use wearable devices (e.g., Fitbit Sense) to continuously monitor temperature and heart rate, then early signs of infection or heart failure will be detected within the first 30 days post-surgery through deviations in these physiological markers.

Rationale:

Cardiovascular complications such as infection or heart failure are most likely to occur within the first 10 days after aortic valve replacement (AVR) surgery, with continued risk for up to 30 days. Traditional monitoring methods, which rely on follow-up visits and patient-reported symptoms, may delay early detection. This study evaluates whether the Fitbit Sense, a commercially available wearable, can serve as a non-invasive tool to continuously monitor heart rate variability (HRV), heart rhythm, and skin temperature during recovery^{vii}. Prior research suggests that small changes in HRV and skin temperature can indicate complications several days before symptoms develop. Algorithms like Night Signal have demonstrated up to 81% sensitivity in identifying such early warning signs. Despite promising data, few studies have applied this technology specifically to AVR recovery. By comparing an experimental group using Fitbit monitoring with a control group receiving standard care, this study aims to determine whether wearables can enable earlier detection, reduce readmission rates, and improve long-term cardiovascular outcomes^{viii}.

Research Methods:

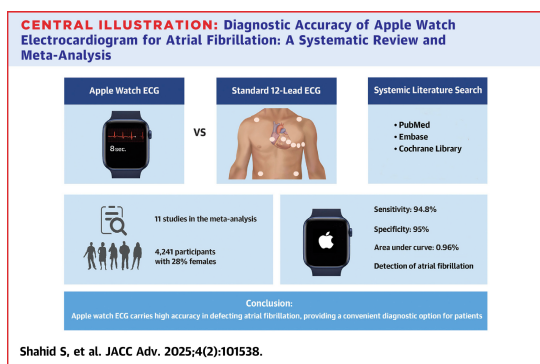


FIGURE 1

This study will use randomly assigned participants to test the hypothesis that continuous physiological monitoring via wearable devices (Fitbit) enables earlier detection of postoperative complications compared to standard care when recovering from aortic valve replacement (A.V.R) surgery. The study will contain 384 adult patients (≥ 18 years) who have undergone A.V.R (excluding those with pre-existing arrhythmias or inability to use wearable technology).

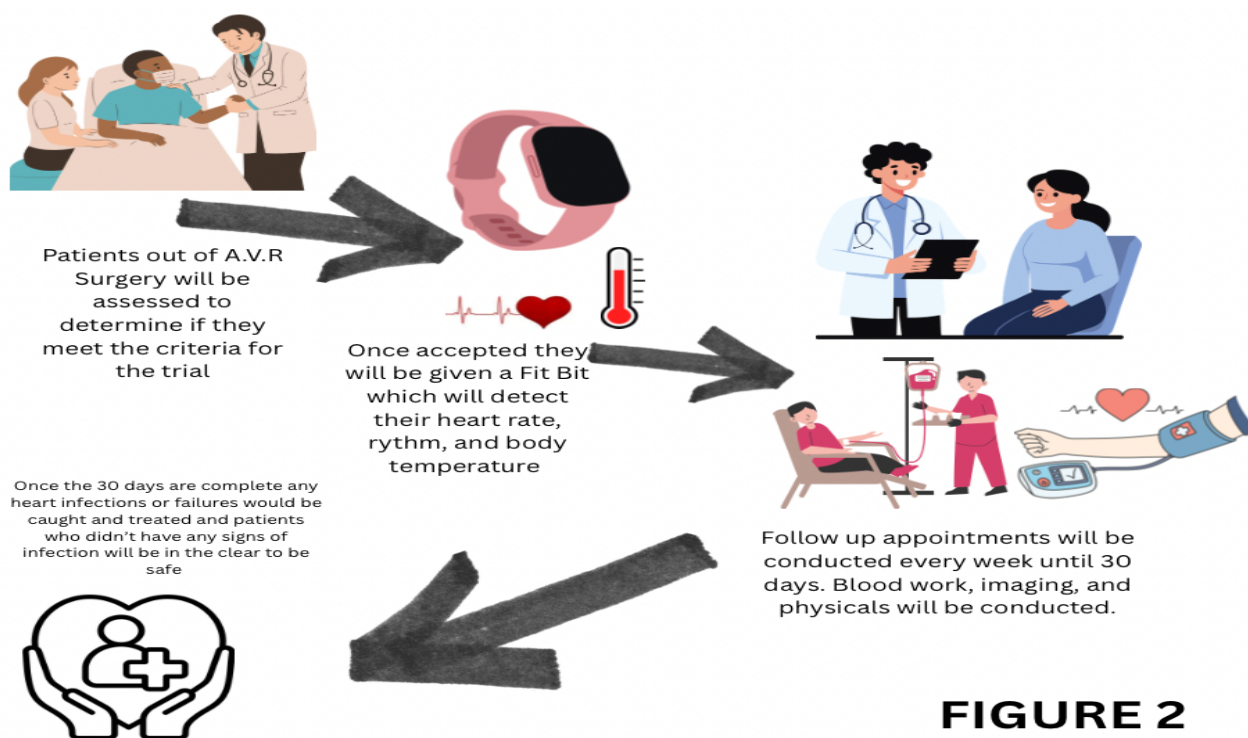


FIGURE 2

Participants will be randomly assigned to either the experimental group (192), which will receive a Fitbit Sense for continuous heart rate, heart rate variability, heart rhythm, and skin temperature monitoring post-discharge, or the control group, which will follow standard postoperative care protocols without wearable monitoring^{ixx}. Prior to surgery, baseline measurements for each patient's heart rate, heart rate variability, heart rhythm, and skin temperature will be recorded over a 24-hour period using the Fitbit Sense^{xi}. The experimental group's Fitbit data will be continuously transmitted to a secure platform where an algorithm will flag deviations from each patient's individual baseline values, triggering clinician alerts for potential complications^{xii}. Both groups will undergo identical scheduled follow-ups every week (4 times) until the final follow up on the 30th day post-surgery, where all suspected infections or heart failure events will be confirmed through standardized diagnostic tests (bloodwork, imaging, clinical assessment ect.).

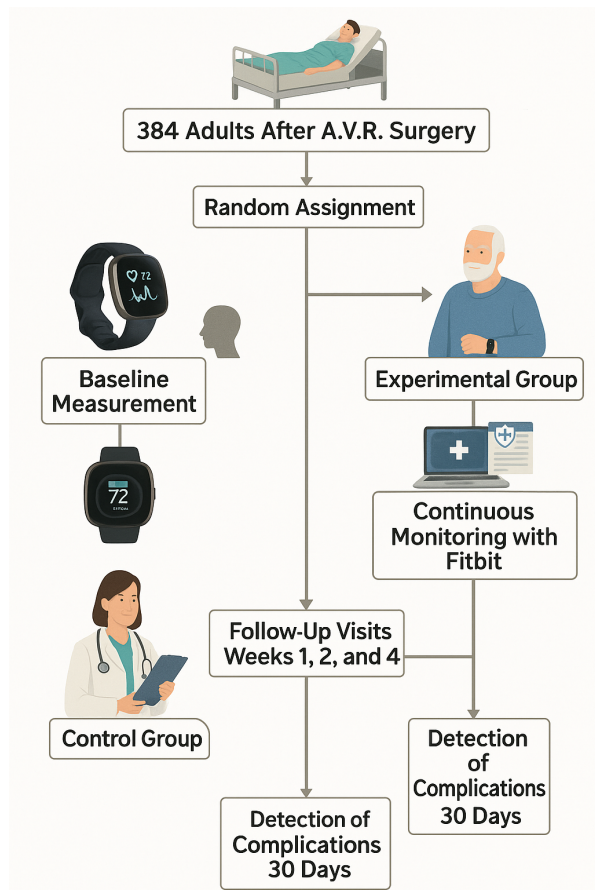


FIGURE 3

The primary outcome is how quickly complications are detected in the wearable group compared to the control group. Secondary measures include hospital readmission rates, alert accuracy, device comfort, and patient adherence. Clinicians will adjust for age and health conditions. The study aims to show whether Fitbit monitoring enables earlier detection and better recovery after surgery^{xiii}.

Conclusion:

This study asks whether a Fitbit can detect early signs of infection or heart failure within 30 days after AVR surgery. The goal is early complication detection, as changes in heart rate and temperature may appear days before symptoms^{xiv}. If Fitbit monitoring proves more effective than standard care, it could improve recovery through real-time alerts. While challenges like proper device use and baseline accuracy exist, the potential benefits are faster treatment and fewer readmissions are significant. Future research may expand to larger, more diverse groups. Ultimately, the study aims to show that wearables offer a simple, non-invasive way to improve post-surgery outcomes^{xv}.

Definition Sheet:

1. **Arrhythmias** – Irregular heart rhythms caused by disruptions in the heart's electrical signals, leading to bradycardia, tachycardia, or fibrillation.
2. **Aortic Valve Replacement (AVR)** – Surgery to replace a damaged aortic valve, often due to narrowing (stenosis) or leakage (regurgitation).
3. **Skin Temperature** – Continuous, non-invasive body temperature monitoring via Fitbit to detect early signs of infection based on changes from baseline.
4. **Night Signal** – A machine learning algorithm that analyzes wearable nighttime data (HR, HRV, skin temp) to detect early signs of infection or heart failure.
5. **Inclusion Criteria** – Conditions required to join the study, such as being 18 or older and having had AVR surgery.
6. **Exclusion Criteria** – Conditions that prevent participation, like having arrhythmias or being unable to use wearable tech.
7. **Secondary Outcomes** – Additional study measures including readmission rates, alert accuracy, device comfort, and patient adherence.

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